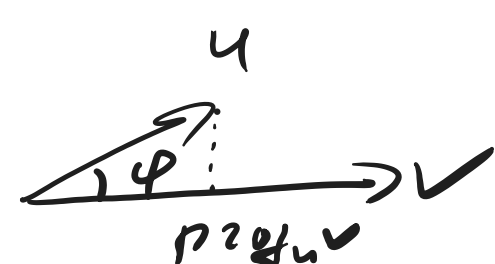


• Math refresher

$$u = \begin{bmatrix} u_0 \\ u_1 \end{bmatrix}$$

$$v = \begin{bmatrix} v_0 \\ v_1 \end{bmatrix}$$



$$\langle u, v \rangle = \text{proj}_u v \cdot |v|$$

$$\langle u, v \rangle = \cos \phi \cdot |u| \cdot |v|$$

$$\begin{pmatrix} - & u & - \end{pmatrix} \begin{bmatrix} 1 \\ v \\ 1 \end{bmatrix} = \begin{pmatrix} u_0 & u_1 \end{pmatrix} \begin{bmatrix} v_0 \\ v_1 \end{bmatrix} = \langle u, v \rangle = u_0 v_0 + u_1 v_1$$

$$A = \begin{bmatrix} - & z_0 & - \\ & z_1 & - \end{bmatrix} ; AB = \begin{bmatrix} - & z_0 & - \\ & z_1 & - \end{bmatrix} \begin{bmatrix} c_0 & c_1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} \langle z_0, c \rangle & \langle z_0, c \rangle \\ \langle z_1, c \rangle & \langle z_1, c \rangle \end{bmatrix}$$

$$B = \begin{bmatrix} c_0 & c_1 \\ 1 & 1 \end{bmatrix}$$

• QKV - Attention

$$\text{softmax} \left(\frac{Q \times K^T}{\sqrt{d_k}} \right) \times V = Z$$

$$\text{softmax}(u) \rightarrow \underline{\underline{0.1777}}$$

$$QK^T = \begin{bmatrix} - & q_0 & - \\ & q_1 & - \end{bmatrix} \begin{bmatrix} k_0 & k_1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -\langle q_0, k \rangle & \langle q_0, k \rangle - \\ \langle q_1, k \rangle & \langle q_1, k \rangle \end{bmatrix}$$

• Masked self-Attn.

$$\begin{cases} \text{softmax} \left(\frac{QK^T + \text{mask}}{\sqrt{d}} \right) V \\ \text{mask} = \begin{bmatrix} 0 & 0 & 0 & -\infty \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{cases}$$